

Harbor View Family Medicine

4422 Pacific Coast Hwy, Suite B, Long Beach, CA 90804
Phone: (562) 555-0271 | Fax: (562) 555-0272

Patient: Eleanor K. Nakamura

Date: 01/20/2026

DOB: 06/15/1994

Provider: Carmen Rodriguez, MD

MRN: HVFM-0041903

Specialty: Family Medicine

CHIEF COMPLAINT

Bilateral leg weakness, right eye vision problem last month, extreme fatigue.

HISTORY OF PRESENT ILLNESS

Ms. Nakamura is a 31-year-old Japanese-American female with no significant past medical history who presents with several weeks of bilateral lower extremity weakness and fatigue. She also reports an episode approximately 6 weeks ago of blurry vision and pain with eye movement in her right eye that lasted about 10 days and then mostly resolved — she did not seek care at that time. Current vision is described as 'almost back to normal' but colors seem slightly washed out in the right eye.

She has had progressive bilateral leg heaviness and weakness over the past 4 weeks, worse in the afternoons and with heat (she noticed it dramatically worsens after a hot shower). She has difficulty climbing stairs and has tripped twice. She is a high school teacher and has been calling in sick frequently. She also reports a band-like tightness around her mid-abdomen that started 3 weeks ago. She has had urinary urgency with two episodes of incontinence, which she finds very distressing.

She denies back pain, recent illness, rash, joint pain, or family history of neurological disease. She is on no medications. She is a non-smoker and drinks socially.

PHYSICAL EXAM

Vitals: BP 118/74, HR 76, Temp 98.3, Wt 138 lbs

General: Alert, appears fatigued, mildly anxious.

Eyes: Visual acuity OD 20/40, OS 20/20. Mild RAPD right eye. Color vision (Ishihara): OD 6/12 plates correct, OS 12/12. Fundoscopy: optic disc OD appears slightly pale — subtle.

Motor: Hip flexors 4/5 bilaterally, knee extensors 4+/5 bilaterally. Ankles 5/5. Upper extremities 5/5.

Reflexes: Bilateral patellar 3+, bilateral ankle 3+. Bilateral Babinski present. Hoffman's bilateral.

Sensation: Vibration mildly reduced bilateral feet. Pinprick intact. Lhermitte's sign: patient reports electric sensation down spine with neck flexion.

Gait: Mildly spastic, circumduction of right leg.

ASSESSMENT AND PLAN

1. Suspected demyelinating disease — probable multiple sclerosis (G35, provisional)

This presentation is very concerning: young woman with subacute bilateral upper motor neuron signs in the legs, prior episode clinically consistent with optic neuritis (right eye pain, vision loss over days, partial recovery, now RAPD and reduced color vision), Lhermitte's sign, Uhthoff's phenomenon (heat sensitivity), urinary symptoms, and sensory level. This represents clinical dissemination in both time and space.

Urgent neurology referral placed — requesting expedited appointment. Ordered MRI brain and cervical spine with and without contrast today (ordered as outpatient but marked urgent). Also ordering CBC, CMP, TSH, B12, ANA, NMO-IgG/MOG antibodies, ESR, CRP, and VEP referral.

Counseled patient that this is serious and requires specialist evaluation. She is understandably frightened. Provided MS Society patient education materials. Told her not to take hot baths.

Pacific Neuroimaging Center

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RADIOLOGY REPORT

Patient: Eleanor K. Nakamura	DOB: 06/15/1994	MRN: HVFM-0041903
Exam Date: 01/22/2026 14:30	Ordering MD: Carmen Rodriguez, MD	Accession: PNC-MR-20260122-3301

EXAM

MRI BRAIN WITH AND WITHOUT GADOLINIUM CONTRAST

CLINICAL INDICATION

31-year-old female with right eye vision loss (partial recovery), bilateral leg weakness, upper motor neuron signs, Lhermitte's sign. Rule out demyelinating disease.

TECHNIQUE

Multiplanar, multisequence MRI brain performed at 3T. Sequences: sagittal T1, axial FLAIR, axial T2, axial DWI/ADC, axial T2*, sagittal MPRAGE pre- and post-contrast, axial T1 post-contrast. Gadavist 0.1 mmol/kg IV administered.

COMPARISON

No prior MRI available.

FINDINGS

White matter: Multiple T2/FLAIR hyperintense lesions are identified, as follows:

- Periventricular: At least 6 ovoid lesions oriented perpendicular to the lateral ventricles (Dawson's fingers appearance), largest measuring 12 x 6 mm in the left periventricular region.
- Juxtacortical: 3 lesions at the gray-white junction, right frontal (2) and left parietal (1).
- Infratentorial: One 5 mm lesion in the right middle cerebellar peduncle.
- Deep white matter: 2 additional lesions in the bilateral centrum semiovale.

Total: At least 12 discrete white matter lesions.

Contrast enhancement: Two lesions demonstrate ring enhancement: one periventricular left (12 x 6 mm) and one juxtacortical right frontal (8 x 4 mm). Enhancement pattern is consistent with active/recent demyelinating plaques (<6 weeks).

Corpus callosum: Multiple T2 lesions along the undersurface of the corpus callosum (callososeptal interface), highly characteristic of MS.

Optic nerves: Right optic nerve shows subtle T2 signal increase and mild enhancement on fat-saturated coronal post-contrast sequence — consistent with resolving optic neuritis.

Gray matter, cortex: No cortical lesions identified on DIR sequence. No atrophy.

Posterior fossa, brainstem: Right MCP lesion as above. No other brainstem lesions.

Ventricles/CSF: Normal. No hydrocephalus.

Vascular: No territorial infarction. No mass effect or midline shift.

IMPRESSION

- Multiple T2/FLAIR white matter lesions in a distribution highly characteristic of multiple sclerosis: periventricular (Dawson's fingers), juxtacortical, infratentorial, and corpus callosum involvement — satisfying MAGNIMS 2016 criteria for dissemination in space.
- Two actively enhancing lesions consistent with acute/subacute demyelinating plaques, indicating recent disease activity.
- Right optic nerve T2 signal change and enhancement consistent with right optic neuritis, correlating with clinical history.
- Findings are consistent with McDonald 2017 criteria for diagnosis of MS when combined with clinical history of prior optic neuritis and current presentation.

5. CSF analysis (for oligoclonal bands) and spinal cord MRI recommended for complete workup.

Electronically attested by: Rachel Kim, MD — Neuroradiology
Read and attested 01/22/2026 18:45 PST | PNC Radiology System

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RADIOLOGY REPORT

Patient: Eleanor K. Nakamura **DOB:** 06/15/1994 **MRN:** HVFM-0041903
Exam Date: 01/22/2026 15:20 **Ordering MD:** Carmen Rodriguez, MD **Accession:** PNC-MR-20260122-3302

EXAM

MRI CERVICAL AND THORACIC SPINE WITH AND WITHOUT GADOLINIUM CONTRAST

CLINICAL INDICATION

Same as MRI brain above. Evaluate for spinal cord demyelinating lesions.

TECHNIQUE

Sagittal T1, T2, STIR, and axial T2 sequences cervical and thoracic spine at 3T. Post-contrast sagittal and axial T1 sequences obtained.

FINDINGS

Cervical spine alignment: Normal lordotic curvature. No fracture or subluxation.

Cervical cord: A single focal T2 hyperintense lesion measuring 8 x 3 mm is identified within the posterior right hemicord at the C4 level. The lesion spans less than 2 vertebral segments in length. No cord expansion or swelling. No abnormal enhancement post-contrast (this lesion is therefore not acutely active). No cord atrophy.

Cervical disc spaces: Mild C5-C6 disc dessication without significant canal or foraminal stenosis. No cord compression.

Thoracic spine: Thoracic cord normal in caliber and signal. One small 4 mm T2 hyperintense lesion in the posterior thoracic cord at T6 level — morphology similar to cervical lesion. No enhancement. No compression.

Vertebral bodies: Normal marrow signal. No compression fractures.

IMPRESSION

- Single right posterior cervical cord lesion at C4 (8 x 3 mm), T2 hyperintense, non-enhancing, less than 2 vertebral segments — morphologically consistent with a non-active demyelinating plaque. Satisfies spinal cord dissemination in space criteria per MAGNIMS 2016.
- Probable additional small demyelinating lesion at T6 level.
- No acute cord compression or myelopathy from structural cause.
- Combined with brain MRI findings, the overall imaging supports a diagnosis of multiple sclerosis with dissemination in space (brain, spinal cord, optic nerve) and in time (enhancing and non-enhancing lesions).

Electronically attested by: Rachel Kim, MD — Neuroradiology
Read and attested 01/22/2026 19:10 PST | PNC Radiology System

Pacific Neuroimaging Center — Neurophysiology Division

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VISUAL EVOKED POTENTIAL (VEP) REPORT

Patient: Eleanor K. Nakamura

DOB: 06/15/1994

MRN: HVFM-0041903

Exam Date: 01/28/2026 10:00

Ordering MD: Sarah Chen, MD (Neurology)

Accession: PNC-VEP-20260128-0214

CLINICAL INDICATION

Suspected right optic neuritis. Prior visual symptoms right eye. Evaluate optic nerve conduction.

TECHNIQUE

Pattern-reversal VEP performed using standard checkerboard stimulus (60' and 15' check sizes) at 2 reversals/second. Electrode placement: Oz, O1, O2, MO, Cz (referential montage). Each eye tested separately with the other eye patched. Corrective lenses worn. Room luminance controlled. 200 averages per trial; trials repeated for reproducibility.

RESULTS

Right Eye (OD):

P100 latency: 126 ms (normal ≤ 115 ms) — PROLONGED

P100 amplitude: 4.2 μ V (low-normal; normal ≥ 4 μ V)

Waveform morphology: Present but broadened.

Left Eye (OS):

P100 latency: 104 ms (normal ≤ 115 ms) — NORMAL

P100 amplitude: 9.8 μ V — Normal

Waveform morphology: Normal.

Interocular latency difference: 22 ms (abnormal; normal ≤ 8 ms)

IMPRESSION

1. ABNORMAL Visual Evoked Potentials.
2. Prolonged P100 latency right eye (126 ms) with significant interocular asymmetry (22 ms difference), consistent with right optic nerve demyelination.
3. Left eye VEP normal.
4. Findings are consistent with prior right optic neuritis with residual conduction delay, correlating with the clinical history and MRI optic nerve findings. This constitutes objective paraclinical evidence supporting the diagnosis of multiple sclerosis.

Electronically attested by: James Park, PhD — Clinical Neurophysiology

Reviewed by: Sarah Chen, MD (Neurology) | Attested 01/28/2026 15:00 PST

UCLA Health Laboratory — Cerebrospinal Fluid Analysis

757 Westwood Plaza, Los Angeles, CA 90095 | CLIA #05D0644312
Phone: (310) 555-0900 | Fax: (310) 555-0901

CEREBROSPINAL FLUID (CSF) ANALYSIS REPORT

Patient: Eleanor K. Nakamura **DOB:** 06/15/1994 **MRN:** HVFM-0041903
Exam Date: 01/29/2026 08:45 **Ordering MD:** Sarah Chen, MD (Neurology) **Accession:** UCLA-CSF-20260129-1109

PROCEDURE NOTE

Lumbar puncture performed at L3-L4 under fluoroscopic guidance by Dr. Chen. Patient in lateral decubitus position. Opening pressure measured. 15 mL CSF collected in sequential numbered tubes. No complications. Patient tolerated procedure well.

OPENING PRESSURE & APPEARANCE

Test	Result	Reference	Flag
Opening Pressure	16 cmH2O	≤20 cmH2O	
Appearance	Clear, colorless	Clear, colorless	
Xanthochromia	Absent	Absent	

CELL COUNTS

Test	Result	Reference	Flag
WBC	3	0-5 cells/μL	
RBC	2	0-5 cells/μL (traumatic tap excluded)	
Differential — Lymphocytes	90%	60-80% of WBC (upper limit)	
Differential — Monocytes	10%	15-45%	
Differential — Neutrophils	0%	0-6%	

CHEMISTRY

Test	Result	Reference	Flag
Glucose (CSF)	62 mg/dL	45-80 mg/dL (serum 94 mg/dL)	
CSF/Serum Glucose Ratio	0.66	≥0.60	
Total Protein	48 mg/dL	15-45 mg/dL	H
Albumin (CSF)	22 mg/dL	10-30 mg/dL	
IgG (CSF)	8.4 mg/dL	0.8-6.0 mg/dL	H
IgG Index	0.82	<0.70 (elevated indicates intrathecal IgG production)	H

OLIGOCLONAL BANDS (Isoelectric Focusing)

Test	Result	Reference	Flag
Serum OCBs	1 band	—	
CSF OCBs	5 bands	—	H
Unique CSF bands (not in serum)	4	0-1 (positive threshold: ≥2 unique bands)	H
Interpretation	POSITIVE — intrathecal IgG synthesis	—	H

OTHER

Test	Result	Reference	Flag
Myelin Basic Protein (MBP)	3.8 ng/mL	<4.0 ng/mL — high-normal	
Culture and Gram Stain	No growth / No organisms	Negative	
VDRL (CSF)	Nonreactive	Nonreactive	
Cytology	No malignant cells	Negative	

INTERPRETATION: CSF profile consistent with intrathecal IgG synthesis: elevated IgG index (0.82) and 4 unique oligoclonal bands not present in serum. These findings, in conjunction with clinical presentation and MRI results, support the diagnosis of multiple sclerosis. Mildly elevated protein and slightly elevated lymphocyte percentage are consistent with active demyelination. No evidence of infection or malignancy.

Electronically attested by: Lisa Wong, MD, PhD — Neuropathology / CSF Laboratory
Authorized 01/29/2026 16:30 PST | UCLA Health Lab